

Tirai — the confidential dealer terminal, on-ledger

A multi-dealer RFQ / OTC desk built native on Canton, settling in real CIP-56 assets (cETH, CBTC, Canton Coin, USDCx).

HackCanton Season #2 · Financial Applications · solo builder: Pugar Huda Mantoro · repo github.com/PugarHuda/tirai · live desk tirai.vercel.app · demo youtu.be/_iHMouFdNA4

THE PROBLEM

When an institution moves a large block of bonds, **the fact of the enquiry is itself the market-moving information**. Ask five dealers for a price on 50 million of a 30-year sovereign and you have told the market your size and your direction before you have traded a single unit.

On a transparent chain this leak is structural, not incidental. An RFQ in a public mempool, a resting order in a public book, a quote visible to rival dealers — each is free information for anyone willing to trade ahead of you. The result is front-running, adverse selection and worse fills. This is precisely why institutional block trading has *not* moved on-chain: it still happens over the phone and on closed dealer terminals (Tradeweb, Bloomberg, MarketAxess), off-ledger, with all the settlement risk and reconciliation cost that implies.

The industry's two answers are both unsatisfying. Trade in public and pay the leak. Or trade off-chain and lose atomic settlement, auditability, and any hope of composing with on-chain assets.

THE VALUE: WHAT TIRAI CHANGES

Tirai is the dealer terminal itself, running on the ledger. A buyer sends an RFQ to a chosen dealer panel; each dealer answers with a **sealed quote**. Rival dealers never receive that contract — not hidden in a UI, *never delivered to their participant node*. The market never sees the RFQ. Losing quotes are archived without ever being revealed.

- **Price discovery without disclosure.** Reverse-Vickrey clears the cheapest ask at the second price, so honest quoting is the dominant strategy; or the buyer lifts a single ask bilaterally. Partial fills on both rails.
- **No counterparty risk.** Quoting escrows the dealer's asset — a price is a commitment, not a bluff — and settlement is atomic delivery-versus-payment: bond and cash move in one transaction, or neither does.
- **Supervision without surveillance.** The regulator observes executed trades only: full post-trade audit, zero pre-trade visibility.
- **Provable best execution.** A buyer (or a dealer defending its pricing) can selectively disclose a single sealed quote to the regulator on demand — proving the clearing price beat every competing ask, without ever going public.

- **The cash leg is real.** Settlement clears in cETH, CBTC, Canton Coin or USDCx through the CIP-56 token standard's allocation flow — one integration, every asset, against the frozen Splice v1 interfaces that live registries implement.

WHY THIS MATTERS NOW

Tokenised fixed income only becomes interesting when institutions can trade it in size. Every other part of the stack — issuance, custody, settlement — has a credible on-chain story; the trading layer does not, because the trading layer is where information leaks. Solve confidential price discovery and the rest of the tokenised bond thesis stops being a pilot.

The same primitive travels: confidential RFQ with atomic DvP is what a cETH or CBTC desk needs to quote size without being picked off, and what an OTC desk in any institutional asset needs on day one.

WHY CANTON, SPECIFICALLY

We have built this exact product four times before — on Arbitrum with trusted execution environments, on Stellar with Groth16 circuits, on Sui with threshold encryption, on Ethereum with fully homomorphic encryption. Every one of those builds was a fight against the chain's transparency, paid for in cryptographic machinery, latency and audit surface.

On Canton we wrote none of it. "Dealer B cannot see dealer A's quote" is a signatory / observer declaration — sub-transaction privacy is the ledger model, not a layer on top. Atomic multi-party DvP across independently administered registries (a bond issuer plus the onRails cETH registry) composes in a single transaction. No other stack gives you confidential pre-trade, provable post-trade and atomic cross-registry settlement together.

PROOF IT EXISTS (LIVE ON CANTON DEVNET)

41 settled trades + 5 atomic multi-leg baskets on-ledger, across reverse-Vickrey, direct-OTC and partial-fill rails, over ~25 sovereign, supranational and corporate instruments.

16 best-execution attestations — clearing price provably at or below every disclosed competing ask.

36 Daml test scripts green; an automated on-ledger check asserts that each dealer sees only its own quotes and the regulator sees zero pre-trade contracts — and fails the build if a single quote ever leaks.

Deployed on HackCanton's own participant too (hackcanton-01, same package id): 20 settled trades + 1 atomic basket and 16 attestations there, privacy verified on that node as well.

Hosted read-only desk at tirai.vercel.app over live Devnet state, plus a 6-tool MCP server so agents can read the desk and post RFQs — and a 4-minute walkthrough at youtu.be/_iHMouFdNA4.

WHO PAYS, AND THE PATH TO PRODUCTION

Customers. Fixed-income and crypto-asset trading desks at banks, asset managers and prop shops that trade in size and cannot afford to signal — plus the venues that would host the desk for them.

Revenue. A per-trade venue fee in the settlement asset (bps of notional), taken atomically at settlement — the economics of an OTC venue, enforced by the contract rather than invoiced. Featured-app activity markers (CIP-0047) accrue network rewards on every settlement, so a live desk keeps earning from the volume it clears.

Next steps. (1) Wire the onRails cETH and BitSafe CBTC registries and run live test-token settlements — the contract path is built and tested, blocked only on the test-token grant. (2) Put the desk in front of one fixed-income and one crypto-native design partner. (3) Supervised pilot at a hosting venue on a single instrument class, with the venue fee and activity markers switched on.

Tirai — Indonesian for "curtain". Price discovery happens behind it. You whisper quotes; the market hears nothing.